Sujan Rada

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.91
Narayana Junior College	Class XII (TSBIE)	2023	98.00%
Narayana Olympiad School	Class X (TS SSC)	2021	10.00%

PROJECTS

Two-Stage CMOS Op-Amp Design with Frequency Compensation :

- Designed a **two-stage Miller-compensated op-amp** in 130 nm CMOS using **xschem** and **ngspice** to meet gain, UGB, phase margin, and slew-rate targets under low supply voltage(**1.2V**).
- Verified stability and performance through ngspice simulations including Bode plots, CMRR, PSRR, transient response, and Total harmonic distortion(THD) analysis.

Analysis and simulation of analog circuit blocks :

- Simulation and mathematical analysis of various architectures of single-stage and differential amplifier architectures using **cadence virtuoso**.
- Implemented multiple architectures of **OP-AMP based oscillators**, **ADC's** and **DAC's** on LTspice and breadboards.
- Implemented **555 timer circuit**s from scratch in monostable, astable, and bistable modes using LTspice.
- Designed and analyzed a block level closed-loop **boost converter** using MATLAB and LTspice.

16X16 DRAM Array Core Read Path Design :

- Modeled **transistor-level schematics** for a **1T-1C DRAM array circuitry**, including memory cell, differential sense amplifier, row decoder, level shifter, pre-charge circuitry and word-line driver in Cadence virtuoso.
- Integrated all the blocks implementing a complete **DRAM core read architecture model** and verified charge sharing, destructive read, sensing, amplification, and write-back mechanisms using transient simulations.
- Devised CMOS decoder and boosted word-line driver for reliable row selection and storage-node restoration.

Transmission Line Impedance Matching Using Smith Chart :

- Analyzed a 50 lossless transmission line to compute , VSWR, return loss, and input impedance using **MATLAB**.
- Designed single-stub shunt matching networks at 3 GHz using Smith chart and analytical validation and evaluated matching bandwidth via frequency sweep and return-loss analysis.

Rotating LED Display :

- Designed a custom **PCB-based rotating LED display** with ESP32-S3, RP2040, and STP24DP05BTR driver ICs using **KiCad** ensuring fully clean electric and design rules check violations (ERC, DRC) for the schematic and PCB layout.
- Integrated a step-down power supply and synchronized LED patterns with rotational speed for Persistence of vision (POV) effects.

EXPERIENCE

Physical Layout design | (IC design Lab - IITH)

- Worked on implementing a **full custom layout for CMOS temperature sensor circuit** in **Cadence Virtuoso**, integrating key analog blocks like OTA, current mirrors, bandgap reference circuit and other digital blocks.
- Used matching techniques like common centroid configuration, dummy transistors placement maintaining proper orientation and spatial symmetry and verified the layout design by ensuring fully clean DRC, LVS and antenna errors.

SKILLS

Programming Languages: Verilog | System Verilog | C | MATLAB | Python | RISCV Assembly | LaTeX

Tools: Cadence Virtuoso | LTSpice | Ngspice | Unix shell (Bash) | Vivado | EDA Playground | GitHub | Altium | KICAD

RELEVANT COURSES

Analog: Circuits and Network Analysis | Semiconductor Device Fundamentals (Device Physics) | Analog Electronics | Analog Lab (Cadence) | Semiconductor Devices Lab (Characterization Lab) | Electronic System Design Project Lab (KiCAD) | PCB Design and Layout for Electronic Systems (Altium Designer) | Waveguides and Antennas | Analog IC Design | Memory Device Technology for AI/ML

POSITIONS OF RESPONSIBILITY

Teaching Assistant - Waveguides and Antennas

Teaching Assistant - Digital Systems Lab

Teaching Assistant - Matrix Theory

Class Representative, (Dept of IC design and technology - IITH)

Hospitality & Logistics Coordinator, Extra Mural Lectures (EML), IITH

UG student mentor, Sunshine (Councelling cell - IITH)

Transport Coordinator, Elan & nVision (Annual techno-cultural Fest - IIT Hyderabad)

EXTRACURRICULARS

- **NSS Volunteer, IIT Hyderabad:** Completed 100+ hours of community service through the National Service Scheme.
- Hobbyist sketcher with an interest in creative arts.
- Active football and water polo player; avid team-sports enthusiast.